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Dr. M. S. Sheshgiri Campus, Belagavi, 590008.

Department of Electronics & Communication Engineering

**A Minor Project Report On
“Real-Time Indoor Localization System Using BLE
Beacon RSSI Fingerprinting and K-Nearest
Neighbor Classification”**

(Semester VI)

(Code: 24EECW302)

Submitted by:

Name	SRN
Parth Jadhav	02FE23BEC083
Md. Danish Shamsher	02FE23BEC112
Roopashree Pote	02FE23BEC122
Keerti Mahantshetti	02FE23BEC123

Under the guidance of:

Prof. Pavan Chavalagi

Department of Electronics & Communication Engineering

KLE Technological University

Dr. M. S. Sheshgiri Campus, Belagavi

Academic Year 2025–2026

CERTIFICATE

This is to certify that the project work entitled “**Real-Time Indoor Localization System Using BLE Beacon RSSI Fingerprinting and K-Nearest Neighbor Classification**” is carried out by **Parth Jadhav** (SRN: 02FE23BEC083), **Md. Danish Shamsher** (SRN: 02FE23BEC112), **Roopashree Pote** (SRN: 02FE23BEC122), and **Keerti Mahantshetti** (SRN: 02FE23BEC123), the bonafide students of **VI Semester** of **KLE Technological University, Dr. M. S. Sheshgiri Campus, Belagavi**, in partial fulfillment for the award of the degree of **Bachelor of Engineering** in the Department of **Electronics and Communication Engineering** during the academic year **2025–2026**.

It is certified that all the corrections and suggestions indicated for internal assessment have been incorporated in the project report deposited in the departmental library. The project report has been approved as it satisfies the academic requirements in respect of the project work prescribed for the said degree.

Guide Signature

Prof. Pavan Chavalagi

HOD Signature

Dr. D. A. Torse

Principal Signature

Dr. S. F. Patil

Examiner – I

Examiner – II

DECLARATION

We hereby declare that the project work presented in this report, entitled “**Real-Time Indoor Localization System Using BLE Beacon RSSI Fingerprinting and K-Nearest Neighbor Classification**”, submitted to KLE Technological University, Dr. M. S. Sheshgiri Campus, Belagavi, for the completion of the course requirements in the VI Semester, is the original work carried out by us in the Department of Electronics and Communication Engineering under the guidance of Prof. Pavan Chavalagi, Department of Electronics and Communication Engineering.

We further declare that, to the best of our knowledge and belief, the work reported herein has not been submitted previously as part of the award of any course, degree, or diploma at this or any other university or institution. The results presented in this report are solely the outcome of our efforts. We also confirm that all the work documented in this report has been completed by us.

Parth Jadhav
SRN: 02FE23BEC083

Md. Danish Shamsher
SRN: 02FE23BEC112

Roopashree Pote
SRN: 02FE23BEC122

Keerti Mahantshetti
SRN: 02FE23BEC123

Date: _____

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— The Project Team

Abstract

Indoor localization is essential in environments where GPS signals are unreliable due to indoor obstructions. This project presents a Real-Time Indoor Localization System using Bluetooth Low Energy (BLE) beacon RSSI fingerprinting and a K-Nearest Neighbor (KNN) classifier implemented with ESP32 devices. RSSI data collected from three BLE beacons is filtered using Median, Weighted Average, and Kalman Filters to reduce noise and improve accuracy. The processed data is used to classify indoor locations such as Room101, Room102, and Corridor in real time. Simulation and testing results achieved an accuracy of 93.42%, demonstrating reliable and efficient indoor positioning suitable for navigation, asset tracking, and smart building applications.

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Chapter 1

Introduction

1.1 Background

Indoor navigation and positioning inside buildings is a major challenge because Global Positioning System (GPS) signals become weak or unavailable due to walls and indoor obstacles. As a result, accurately determining the location of a person or device inside hospitals, shopping malls, airports, and smart buildings becomes difficult. This limitation affects indoor tracking, navigation, and location-based services that require precise and reliable positioning.

With the rapid growth of Internet of Things (IoT) devices and smart infrastructure, demand for low-cost and scalable indoor positioning solutions has increased significantly. Bluetooth Low Energy (BLE) technology has emerged as one of the most practical approaches for indoor localization due to its low power consumption, low hardware cost, and widespread availability in consumer devices.

1.2 Problem Statement

The primary problem addressed in this project is the design of a low-cost and reliable indoor human localization system using Bluetooth Low Energy (BLE) signals and machine learning techniques. The system continuously collects RSSI (Received Signal Strength Indicator) values from multiple BLE beacons placed at fixed indoor locations and predicts the user's indoor position in real time using a trained KNN classifier.

GPS-based localization fails indoors due to:

- Signal attenuation caused by walls and building materials
- Multipath propagation and signal reflection

- Lack of direct satellite visibility inside enclosed structures
- High infrastructure and deployment cost for alternative solutions

The proposed system addresses these limitations using low-cost ESP32 devices, BLE-based RSSI fingerprinting, signal preprocessing filters, and machine learning classification to achieve reliable and real-time indoor positioning.

1.3 Objectives

The main objectives of this project are:

- To develop a low-cost indoor localization system using BLE technology and ESP32 devices.
- To collect RSSI measurements from multiple BLE beacons placed at known indoor locations.
- To reduce RSSI fluctuations and signal noise using Median Filter, Weighted Average, and Kalman Filter techniques.
- To train and evaluate machine learning models for accurate indoor location classification.
- To compare the performance of KNN, SVM, Random Forest, and Decision Tree classifiers.
- To demonstrate real-time location prediction and monitoring through a Python-based interface.

1.4 Scope of the Project

The scope of this project covers room-level indoor localization across three zones: Room101, Room102, and Corridor. The system uses three fixed ESP32 BLE beacons and one ESP32 scanner node. RSSI-based fingerprinting is used to build a training dataset, and the KNN classifier is used for real-time prediction. The system is validated through simulation on the Wokwi platform and tested using Python-based prediction scripts.

Chapter 2

Literature Survey

Indoor localization has been an active area of research due to the limitations of GPS-based positioning systems in enclosed environments. Several technologies and approaches have been proposed including Wi-Fi RSSI fingerprinting, UWB-based ranging, infrared sensors, and BLE beacon-based systems. This chapter reviews relevant prior work in the domain and identifies the research gap addressed by the proposed system.

2.1 GPS Limitations Indoors

GPS technology provides accurate outdoor positioning but performs poorly indoors due to signal attenuation caused by walls, ceilings, and furniture. Singh et al. (2021) demonstrated that GPS signals experience severe multipath interference and signal blockage inside buildings, making GPS unsuitable for indoor localization without additional infrastructure [1].

2.2 BLE-Based Indoor Localization

Bluetooth Low Energy has emerged as a preferred technology for indoor localization due to its low power consumption, low hardware cost, and widespread deployment capability. Cortesi et al. (2023) proposed a BLE fingerprinting system using multiple beacon nodes for room-level classification and demonstrated accuracy above 90% in controlled environments [2]. Milano et al. (2024) further explored BLE-based systems integrated with machine learning algorithms and reported improved localization stability through RSSI preprocessing [5].

2.3 RSSI Signal Challenges

RSSI-based localization is affected by several environmental factors including multipath propagation, signal interference from electronic devices, attenuation due to obstacles, and human body blockage. Tilwari et al. (2024) analyzed RSSI variation patterns in indoor environments and proposed preprocessing techniques to stabilize signal readings before classification [3]. Ghaemifar et al. (2025) highlighted the importance of Kalman filtering in reducing RSSI noise for improved localization accuracy [6].

2.4 Machine Learning for Indoor Localization

Machine learning techniques have been widely applied to indoor localization using RSSI fingerprinting. Bouse et al. (2024) compared KNN, SVM, and Random Forest classifiers for BLE-based localization and concluded that KNN provided consistent accuracy with lower computational overhead compared to SVM [4]. Yang et al. (2025) explored deep learning models for indoor positioning and reported improved accuracy for large-scale environments while noting higher computational requirements [7].

2.5 Filtering Techniques for RSSI Preprocessing

Skýpalová et al. (2025) studied different signal filtering methods for RSSI stabilization and reported that combining Median Filter with Kalman Filter significantly reduced signal variance and improved classifier performance [8]. Swargam et al. (2023) demonstrated that weighted averaging of RSSI samples provided better temporal stability compared to raw RSSI measurements [9].

2.6 Research Gap

From the literature survey, it is observed that many existing indoor localization systems require expensive hardware infrastructure, large training datasets, or high computational resources. Limited work has focused on combining low-cost ESP32-based BLE beacons with multi-stage RSSI filtering and lightweight KNN classification for real-time indoor localization. The proposed system addresses this gap by providing a fully integrated, low-cost, and deployable solution using BLE RSSI fingerprinting, signal preprocessing, and machine learning classification.

Chapter 3

System Architecture and Methodology

3.1 Proposed System Overview

The proposed system implements a real-time indoor localization framework using three ESP32 BLE beacon nodes and one ESP32 scanner node. The scanner collects RSSI values from all three beacons, applies a multi-stage signal preprocessing pipeline, and transmits the filtered RSSI values to a Python-based classification engine. The KNN classifier then predicts the indoor location in real time.

3.2 Block Diagram

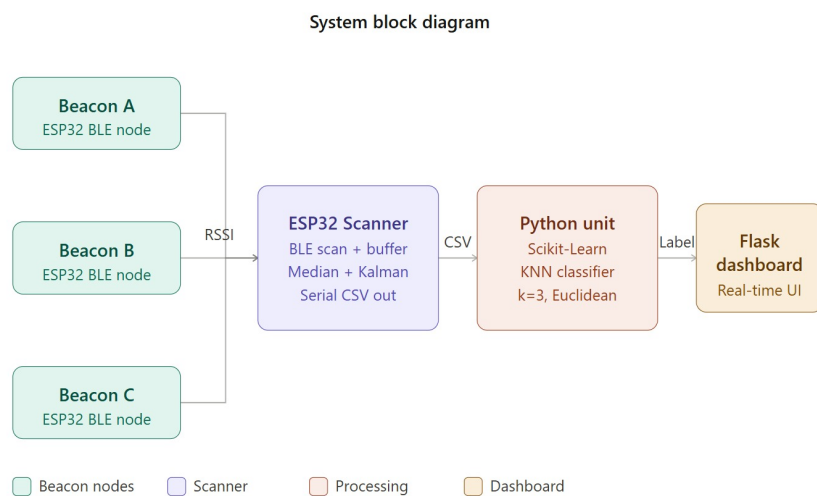


Figure 3.1: Proposed Indoor Localization System Block Diagram

The system consists of the following major components:

- **BLE Beacon Nodes (Beacon A, B, C):** Fixed ESP32 devices that continuously broadcast BLE advertisement packets.
- **ESP32 Scanner Node:** Receives BLE advertisements, extracts RSSI values, and applies preprocessing filters.
- **Python Processing Unit:** Receives filtered RSSI data over Serial, runs the KNN classifier, and outputs the predicted location.
- **Real-Time Output:** Displays the predicted location and confidence score.

3.3 System Architecture

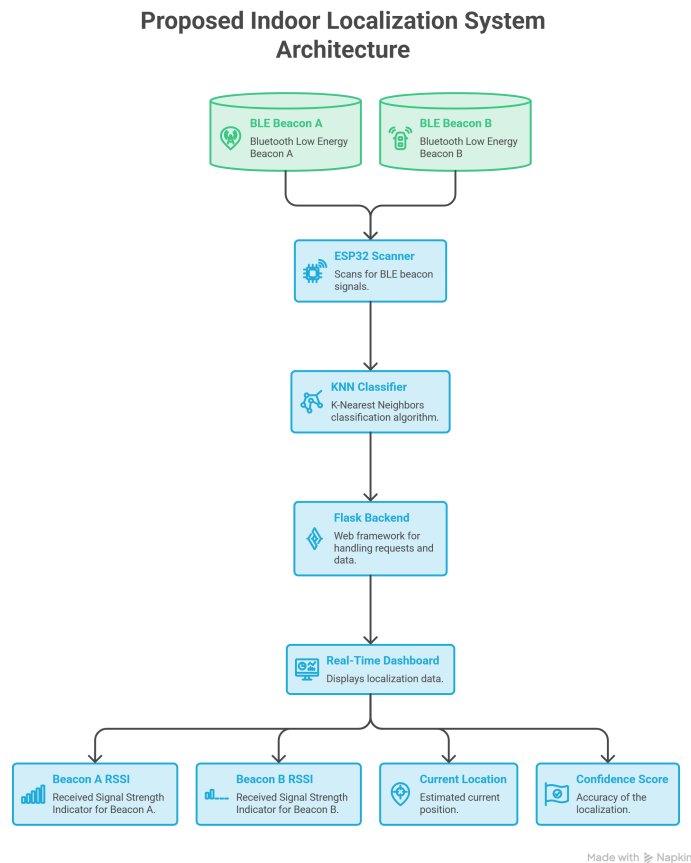


Figure 3.2: Proposed Indoor Localization System Architecture

3.4 Hardware Components

3.4.1 ESP32 BLE Beacon

Three ESP32 devices are configured as BLE beacon nodes. Each beacon continuously broadcasts BLE advertisement packets with a unique device name (BeaconA, BeaconB,

BeaconC). The beacons are placed at fixed known positions inside the indoor environment to create unique RSSI fingerprints for each location zone.

3.4.2 ESP32 Scanner

One ESP32 device is configured as the BLE scanner. The scanner performs active BLE scanning, detects all three beacons, and extracts RSSI values. The RSSI values are stored in circular buffers of size 10 samples per beacon. The buffered values are then processed through the filtering pipeline before being transmitted over Serial.

3.5 Software Components

3.5.1 Arduino IDE

The ESP32 firmware for both beacon and scanner nodes is developed using Arduino IDE. The beacon code handles BLE advertising while the scanner code handles BLE scanning, RSSI buffering, filtering, and Serial transmission.

3.5.2 Python Classification Engine

The Python script receives filtered RSSI values from the ESP32 scanner over Serial, formats them as a feature vector, and passes them to the trained KNN model for real-time location prediction.

3.5.3 Wokwi Simulation

The Wokwi platform is used for hardware simulation prior to physical implementation. The BLE beacon and scanner behavior is verified on Wokwi to ensure correct RSSI collection and Serial output.

3.6 Working Principle

The system operates in the following steps:

1. BLE beacons broadcast advertisement packets continuously.
2. The ESP32 scanner detects all three beacons and extracts RSSI values.
3. RSSI values are stored in circular buffers (size = 10 samples per beacon).
4. Median filtering is applied to remove outlier samples.

5. Weighted averaging is applied to prioritize recent RSSI measurements.
6. The Kalman filter is applied to smooth the RSSI signal over time.
7. The final filtered RSSI values are transmitted via Serial in CSV format.
8. Python receives the RSSI values and passes them to the KNN classifier.
9. The KNN model predicts the indoor location with a confidence score.

Chapter 4

Dataset Collection and Preprocessing

4.1 Dataset Collection

The dataset for training the KNN classifier was collected by placing the ESP32 scanner at different indoor locations and recording the RSSI values from all three beacons. Data was collected for three indoor zones: Room101, Room102, and Corridor. Multiple RSSI readings were collected at each zone under different environmental conditions to ensure dataset diversity and robustness.

The dataset features are:

- Beacon A RSSI (dBm)
- Beacon B RSSI (dBm)
- Beacon C RSSI (dBm)

The dataset labels are:

- Room101
- Room102
- Corridor

The dataset was split into 80% training and 20% testing subsets for model evaluation.

4.2 RSSI Signal Preprocessing

Raw RSSI values exhibit significant fluctuations due to multipath propagation, interference, and environmental changes. A three-stage preprocessing pipeline is applied to stabilize RSSI values before classification.

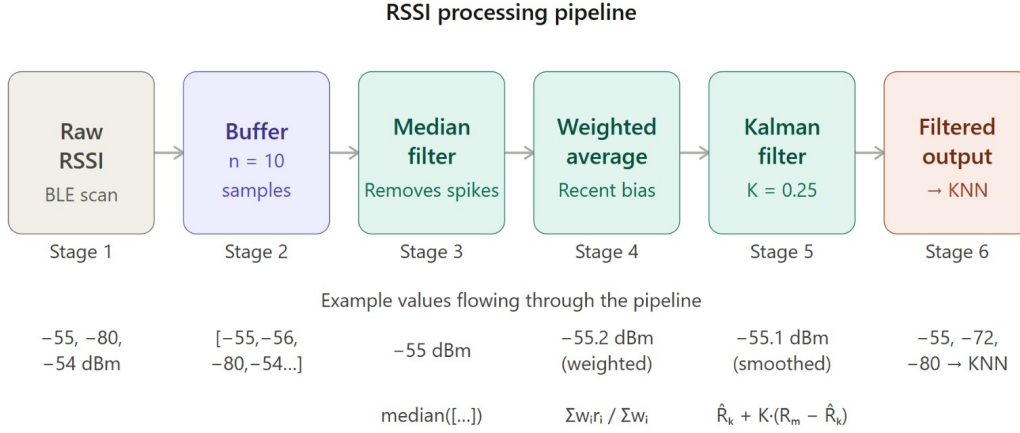


Figure 4.1: RSSI Signal Preprocessing Pipeline

The Median Filter is applied first to remove outlier RSSI samples from the circular buffer. The median value of the last 10 samples is taken as the representative RSSI value for each beacon. This eliminates sudden spikes or drops in RSSI caused by temporary interference.

For a buffer of n samples $[r_1, r_2, \dots, r_n]$, the median filter output is:

$$RSSI_{median} = \text{median}(r_1, r_2, \dots, r_n) \quad (4.1)$$

4.2.1 Weighted Average

A Weighted Average is applied to prioritize more recent RSSI measurements over older ones. Weights increase linearly from older to newer samples, ensuring that the most recent signal readings have a greater influence on the final RSSI estimate.

$$RSSI_{WA} = \frac{\sum_{i=1}^n w_i \cdot r_i}{\sum_{i=1}^n w_i} \quad (4.2)$$

where w_i is the weight assigned to sample r_i and n is the total number of samples in the buffer.

4.2.2 Kalman Filter

The Kalman Filter is applied as the final stage of preprocessing to smooth the RSSI signal over time and reduce random noise. The Kalman gain K is set to 0.25, which balances noise reduction and responsiveness to real signal changes.

$$RSSI_k = RSSI_{k-1} + K \cdot (RSSI_{measured} - RSSI_{k-1}) \quad (4.3)$$

where $RSSI_k$ is the filtered output at time k , $RSSI_{k-1}$ is the previous estimate, $RSSI_{measured}$ is the current raw measurement, and $K = 0.25$ is the Kalman gain.

4.3 Serial Data Transmission

After preprocessing, the ESP32 scanner transmits the filtered RSSI values to the Python host via Serial in the following CSV format:

CSV: -55, -72, -80

The Python script parses this CSV string, extracts the three RSSI values, and passes them as a feature vector to the KNN classifier.

Chapter 5

Machine Learning Model Development

5.1 Overview

The machine learning component of the system is responsible for mapping filtered RSSI feature vectors to indoor location labels. Four classifiers were evaluated: K-Nearest Neighbors (KNN), Support Vector Machine (SVM), Random Forest, and Decision Tree. All models were trained on the same 80% training split and evaluated on the 20% test split.

5.2 K-Nearest Neighbors Classifier

K-Nearest Neighbors is a non-parametric, instance-based learning algorithm that classifies a new data point based on the majority label among its k nearest neighbors in the feature space. For this system, $k = 3$ and the Euclidean distance metric is used.

The Euclidean distance between a query point x and a training sample y in m -dimensional feature space is given by:

$$d(x, y) = \sqrt{\sum_{i=1}^m (x_i - y_i)^2} \quad (5.1)$$

where x_i and y_i are the i -th feature values of the query point and training sample respectively, and $m = 3$ (Beacon A, B, C RSSI values).

The KNN classifier identifies the 3 nearest training samples to the query RSSI vector

and assigns the most frequent label among them as the predicted location.

5.3 Real-Time Prediction Pipeline

The real-time prediction pipeline operates as follows:

1. ESP32 scanner transmits filtered RSSI values over Serial.
2. Python reads the Serial stream and parses the CSV string.
3. The RSSI feature vector is passed to the trained KNN model.
4. KNN predicts the location label and computes the confidence score.
5. The predicted location and confidence are displayed in real time.

Example output:

```
Location: Room101    Confidence: 100%
```

5.4 Model Selection and Comparison

All four classifiers were trained and evaluated under identical conditions to ensure a fair comparison. Table 5.1 summarizes the accuracy results.

Table 5.1: Machine Learning Model Accuracy Comparison

Model	Accuracy (%)
K-Nearest Neighbors (KNN)	93.42
Support Vector Machine (SVM)	93.86
Random Forest	92.98
Decision Tree	92.98

Although SVM achieved the highest accuracy of 93.86%, KNN was selected as the final classifier for the following reasons:

- **Similar Accuracy:** The accuracy difference between KNN and SVM is only 0.44%, which is negligible for this application.
- **Faster Training:** KNN requires no explicit training phase. The training time was measured at approximately 0.0021 seconds.

- **Lightweight:** KNN has a simple architecture with minimal memory requirements, making it suitable for embedded and edge computing deployments.
- **Easy Integration:** The KNN model can be directly integrated with the ESP32 Serial pipeline without complex preprocessing of model parameters.
- **Explainability:** KNN predictions are easily explainable and interpretable, which is important for system validation and debugging.
- **RSSI Fingerprinting Compatibility:** RSSI fingerprinting naturally aligns with the nearest-neighbor search principle of KNN, making it the most intuitive classifier for this problem.

Chapter 6

Results and Discussion

6.1 Filtering Effectiveness

The three-stage RSSI preprocessing pipeline significantly improved signal stability compared to raw RSSI measurements. The Median Filter successfully removed outlier samples caused by sudden interference. The Weighted Average ensured recent measurements contributed more to the RSSI estimate. The Kalman Filter provided smooth and continuous RSSI tracking over time. The combined effect of all three filters resulted in stable and reliable RSSI features for classification.

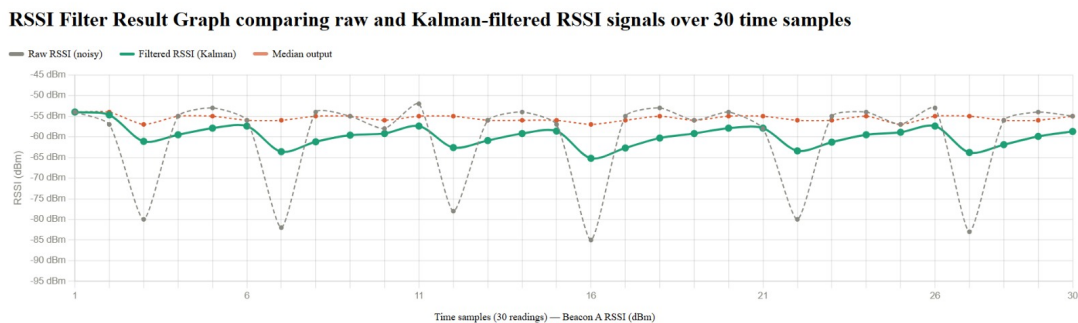


Figure 6.1: RSSI Signal Before and After Filtering

6.2 Real-Time Localization Testing

Real-time localization tests were conducted by placing the scanner at known indoor positions and observing the predicted output. Table 6.1 summarizes representative test cases with input RSSI vectors, predicted locations, and confidence scores.

Table 6.1: Real-Time Localization Test Results

Beacon A (dBm)	Beacon B (dBm)	Beacon C (dBm)	Predicted Location	Confidence
-52	-75	-79	Room101	100%
-73	-52	-77	Room102	100%
-63	-64	-52	Corridor	67.57%

The system correctly predicted Room101 and Room102 with 100% confidence in the test cases. Corridor prediction showed slightly lower confidence of 67.57%, which is expected since corridor environments have more variable RSSI patterns due to open space and signal overlap from adjacent rooms.

6.3 Simulation Results

```

4 from sklearn.neighbors import KNeighborsClassifier
5 from sklearn.model_selection import train_test_split
6 from sklearn.metrics import accuracy_score
7
8 # Training Data
9
10 data_A = [
11     # Room101 samples
12     -52, -51, -53, -50, -52, -54, -51, -53, -52, -50,
13     # Room102 samples
14     -75, -76, -74, -77, -75, -73, -76, -74, -75, -76,
15     # Corridor samples
16     -63, -64, -62, -65, -63, -61, -64, -62, -63, -64,
17 ]
18
19 data_B = [
20     # Room101
21     -76, -75, -77, -74, -76, -78, -75, -77, -76, -74,
22     # Room102
23     -51, -52, -50, -53, -51, -49, -52, -50, -51, -52,
24     # Corridor
25     -62, -63, -61, -64, -62, -60, -63, -61, -62, -63,
26 ]
27
28 labels = ["Room101"] * 10 + ["Room102"] * 10 + ["Corridor"] * 10
29
30 X = list(zip(data_A, data_B))
31 y = labels

```

```

[ESP32]
[ESP32] BeaconA Raw: -60 Median: -60 WAvg: -60 Final: -59
[ESP32] BeaconB Raw: -88 Median: -87 WAvg: -87 Final: -84
[ESP32] BeaconA: -59 dBm | BeaconB: -84 dBm
[ESP32] Location : Room101
[ESP32] Confidence : 100.0%
[ESP32] All scores : Corridor: 0% Room101: 100% Room102: 0%

```

Figure 6.2: Simulation of ESP32 BLE Scanner

6.4 Python Dashboard Output

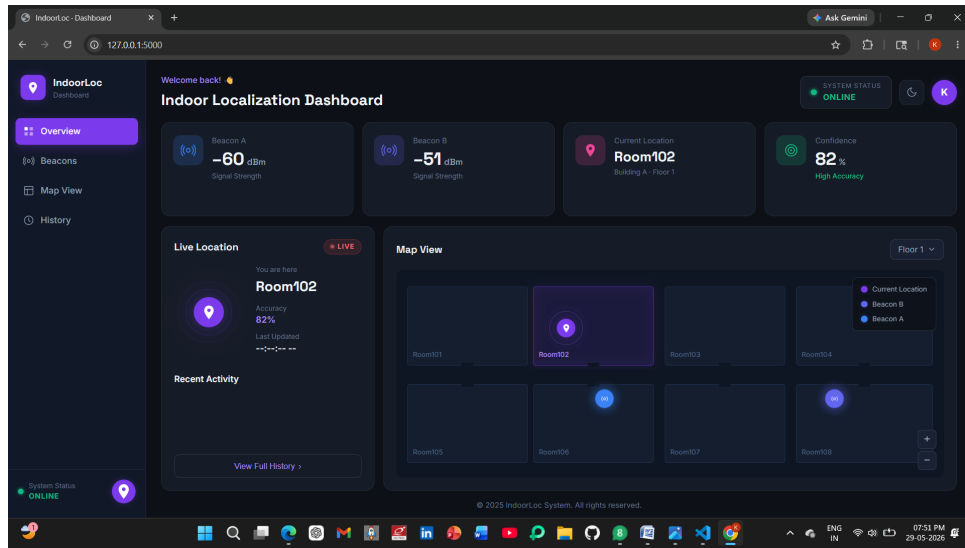


Figure 6.3: Real-Time Location Prediction Output from Python

6.5 Performance Analysis

The proposed system demonstrates reliable room-level indoor localization using low-cost ESP32 hardware. The KNN classifier achieved 93.42% accuracy on the test dataset. The preprocessing pipeline improved RSSI stability and contributed to consistent classification results. Table 6.2 summarizes the key performance metrics of the proposed system.

Table 6.2: System Performance Summary

Parameter	Value
Number of BLE Beacons	3
Number of Location Zones	3
KNN Accuracy	93.42%
Dataset Split (Train/Test)	80% / 20%
KNN Parameter (k)	3
Distance Metric	Euclidean
Kalman Gain (K)	0.25
Buffer Size	10 samples
Real-Time Prediction	Yes
Hardware Cost	Low (ESP32 only)

Chapter 7

Conclusion

This project successfully developed a Real-Time Indoor Localization System using BLE Beacon RSSI Fingerprinting and the K-Nearest Neighbor (KNN) algorithm with ESP32 devices. By applying RSSI preprocessing techniques such as Median Filter, Weighted Average, and Kalman Filter, the system effectively reduced signal noise and achieved reliable indoor location prediction. The KNN classifier attained an accuracy of 93.42%, demonstrating that BLE-based localization can provide an accurate, low-cost, and practical solution for indoor navigation, asset tracking, and smart building applications.

7.1 Future Scope

The system can be further enhanced by deploying additional BLE beacons to improve localization accuracy and coverage. Future work may also include implementing coordinate-based localization techniques such as trilateration, exploring advanced deep learning models, integrating cloud-based IoT monitoring platforms, and developing mobile applications for real-time location visualization. Furthermore, combining BLE with technologies such as Wi-Fi, UWB, and IMU sensors through sensor fusion can improve positioning accuracy and enable large-scale deployment in environments such as hospitals, shopping malls, and airports.

Bibliography

- [1] N. Singh, S. Choe, and R. Punmiya, “Machine Learning Based Indoor Localization Using Wi-Fi RSSI Fingerprints: An Overview,” *IEEE Access*, vol. 9, pp. 127150–127174, 2021.
- [2] S. Cortesi, M. Dreher, and M. Magno, “Design and Implementation of an RSSI-Based Bluetooth Low Energy Indoor Localization System,” 2023.
- [3] V. Tilwari, S. Pack, M. W. P. Maduranga, and H. K. I. S. Lakmal, “An Improved Wi-Fi RSSI-Based Indoor Localization Approach Using Deep Randomized Neural Network,” *IEEE Transactions on Vehicular Technology*, vol. 73, no. 12, pp. 18593–18604, 2024.
- [4] L. Bouse, J. Kral, and M. Hromada, “Simplified Indoor Localization Using Bluetooth Beacons and RSS Fingerprinting,” *Sensors*, vol. 24, no. 7, 2024.
- [5] F. Milano, A. Narduzzi, and G. Pasolini, “BLE-Based Indoor Localization: Analysis of Some Solutions for Performance Improvement,” *Sensors*, vol. 24, no. 2, 2024.
- [6] M. Ghaemifar *et al.*, “Bluetooth Low Energy for Indoor Positioning: Challenges and Current Trends,” *Journal of Ambient Intelligence and Humanized Computing*, 2025.
- [7] Y. Yang *et al.*, “A Bluetooth Indoor Positioning System Based on Deep Learning,” *Sensors*, vol. 25, no. 9, 2025.
- [8] E. Skýpalová *et al.*, “Innovative Indoor Positioning Using Bluetooth Low Energy Beacons,” *Electronics*, vol. 14, no. 10, 2025.
- [9] B. K. Swargam, A. K. Sahoo, and S. S. Dash, “Two Level Wi-Fi Fingerprinting Based Indoor Localization Using Machine Learning,” in *Proc. International Conference on Computing and Network Communications*, 2023.
- [10] B. Shin, J. Kim, and H. Kim, “Novel Indoor Fingerprinting Method Based on RSS Vector Sequence,” *Measurement*, vol. 220, 2023.